

CCSU BCA Semester 2 – Computer Programming / Problem Solving

2026 Exam-Oriented Important Topics + Short Notes

1. Algorithm & Flowchart

Short Note: Algorithm is a finite step-by-step solution. Learn characteristics, flowchart symbols and tracing.

Exam focus: Definition + key points + example/diagram/program where applicable.

2. Variables & Data Types

Short Note: Learn identifiers, constants, primitive data types, input/output and type conversion.

Exam focus: Definition + key points + example/diagram/program where applicable.

3. Operators & Expressions

Short Note: Practice arithmetic, relational, logical, assignment and conditional operators, including precedence.

Exam focus: Definition + key points + example/diagram/program where applicable.

4. Control Statements

Short Note: Prepare if/else, switch, for, while and do-while with simple programs.

Exam focus: Definition + key points + example/diagram/program where applicable.

5. Functions

Short Note: Learn declaration, definition, parameters, return values, scope and recursion.

Exam focus: Definition + key points + example/diagram/program where applicable.

6. Arrays & Strings

Short Note: Practice one/two-dimensional arrays, searching, sorting and basic string operations.

Exam focus: Definition + key points + example/diagram/program where applicable.

7. Important Programs

Short Note: Prepare factorial, prime, palindrome, Fibonacci, reverse number, GCD, largest number and basic array programs.

Exam focus: Definition + key points + example/diagram/program where applicable.

■ Important Questions

- Explain **Algorithm & Flowchart** with suitable example/diagram/formula/program.
- Explain **Variables & Data Types** with suitable example/diagram/formula/program.
- Explain **Operators & Expressions** with suitable example/diagram/formula/program.
- Explain **Control Statements** with suitable example/diagram/formula/program.
- Explain **Functions** with suitable example/diagram/formula/program.
- Explain **Arrays & Strings** with suitable example/diagram/formula/program.
- Explain **Important Programs** with suitable example/diagram/formula/program.

Note: These are revision notes aligned to the CCSU BCA curriculum structure available for 2025–26. Check your college's paper combination/electives before the exam.